Climate Change Knowledge Graph

End-to-End Pipeline: Crawling, KB Construction, Reasoning, KGE and RAG

Web Datamining and Semantics : Final Project

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Table of Contents

1. Data Acquisition and Information Extraction 3
   1.1 Domain and Seed URL Selection 3
   1.2 Crawler Design and Ethics 3
   1.3 NER Results 3
   1.4 Relation Extraction 4
   1.5 Ambiguity Analysis 4
2. Knowledge Base Construction, Alignment and Expansion 5
   2.1 RDF Modeling Choices 5
   2.2 Entity Linking with Confidence 5
   2.3 Predicate Alignment 6
   2.4 Expansion Strategy 6
   2.5 Final KB Statistics 6
3. Reasoning with SWRL 7
   3.1 SWRL Rule on family.owl 7
   3.2 SWRL Rule on the Climate Change KB 7
4. Knowledge Graph Embeddings 8
   4.1 Data Preparation 8
   4.2 Model Comparison: TransE vs DistMult 8
   4.3 Size Sensitivity Analysis 8
   4.4 t-SNE Visualization of Entity Embeddings 9
5. RAG over RDF/SPARQL 10
   5.1 Setup 10
   5.2 Schema Summary 10
   5.3 NL to SPARQL Prompt Template 10
   5.4 Self-Repair Mechanism 10
   5.5 Evaluation 10
   5.6 Demo Screenshots 11
6. Critical Reflection 13
   6.1 KB Quality Impact 13
   6.2 Noise Issues 13
   6.3 Rule-Based vs Embedding-Based Reasoning 13
   6.4 What We Would Improve 13

Pipeline Overview

The following diagram summarizes the end-to-end pipeline implemented in this project, from raw web content to the final RAG demo.

1. Data Acquisition and Information Extraction

1.1 Domain and Seed URL Selection

The chosen domain is Climate Change, a knowledge-rich field with dense entity coverage across organizations (UN bodies, research institutes), geopolitical entities (countries, regions), persons (scientists, policymakers), and temporal references (treaties, milestones). This domain is well-represented in Wikidata and DBpedia, which is critical for the alignment and expansion steps.

Three seed URLs were retained from authoritative sources after filtering:

- https://www.un.org/en/global-issues/climate-change (1,560 words)
- https://science.nasa.gov/climate-change/effects/ (1,468 words)
- https://climate.nasa.gov/evidence/ (1,721 words)

Three additional URLs (NOAA, World Bank, UNFCCC) were discarded due to failed extraction or content below the 500-word threshold.

1.2 Crawler Design and Ethics

The pipeline was implemented in Python using trafilatura for HTML fetching and boilerplate removal, and spaCy (en_core_web_trf, transformer-based) for NLP processing. The extraction pipeline followed these steps: (1) fetch raw HTML via trafilatura.fetch_url(), (2) extract main content via trafilatura.extract() which strips navigation menus, footers, and advertisements, (3) filter pages with fewer than 500 words, and (4) store valid pages in JSONL format preserving the URL-to-text mapping.

The crawler operates ethically by targeting a small set of well-known institutional URLs, avoiding aggressive crawling, and respecting the implicit rate limits of institutional servers. No robots.txt violations occurred given the manual seed selection approach.

1.3 NER Results

Label	Count	Percentage
DATE	59	22.1%
ORG	69	25.8%
GPE	29	10.9%
PERSON	110	41.2%
TOTAL	267	100%

A total of 267 entities were extracted across the three retained pages using spaCy en_core_web_trf. PERSON entities dominate due to the dense citation content of the NASA pages. DATE entities represent a smaller proportion than with the base model, reflecting the transformer model's more precise temporal discrimination.

1.4 Relation Extraction

Candidate relations were extracted by identifying sentences containing at least two entities of the target types. For each such sentence, all entity pairs were recorded as candidate triples (subject, relation, object). The relation was extracted as the lemma of the first verb in the sentence using spaCy's POS tagger; when no verb was found, the relation defaulted to co_occurrence. A total of 1,313 candidate relations were extracted. This heuristic is intentionally lightweight, prioritizing coverage over precision. A more precise approach using spaCy's dependency parser (nsubj and dobj tags) would yield higher-quality relations but at the cost of reduced recall.

1.5 Ambiguity Analysis

Three representative cases of entity ambiguity were identified in the extraction output.

Case 1: Bibliographic authors tagged as PERSON. Names such as Stocker, D. Qin, and G.-K. Plattner appear in bibliographic references at the bottom of the NASA page and are correctly identified as person names by spaCy. In a Knowledge Graph context, they are citation metadata rather than domain entities. Mitigation: filter entities appearing exclusively within bibliographic sections, or apply a heuristic to exclude initials-based name patterns.

Case 2: United Nations tagged as GPE instead of ORG. The model assigns GPE to United Nations because it superficially resembles a country-level entity in its training data. This is a systematic error affecting every occurrence of the phrase. Mitigation: apply a post-processing rule to reclassify known organization names from GPE to ORG.

Case 3: Vague temporal expressions tagged as DATE. Expressions such as today, this decade, and the next few years are classified as DATE by spaCy despite carrying no concrete date value. Mitigation: filter DATE entities to retain only those matching four-digit year patterns or standardized date formats.

2. Knowledge Base Construction, Alignment and Expansion

2.1 RDF Modeling Choices

The initial private Knowledge Base was constructed from the extracted entities and candidate relations using rdflib. Entities were typed according to four classes defined in the private namespace http://climate-change.org/kg/: Person, Organization, GeopoliticalEntity, and Date. Each entity was assigned a URI derived from its label text and annotated with rdfs:label. Relations from the candidate_relations.csv file were added as typed predicates between entities that both existed in the graph.

The ontology (ontology.ttl) defines six OWL classes, four object properties with domain and range constraints, and two datatype properties. Two additional classes were introduced beyond the NER labels: ClimateEvent and Agreement (a subclass of Organization). Key object properties include affiliatedWith, locatedIn, signedBy, and occursIn.

Component	Value
Initial entities	212
Initial relations	1,199
Initial triples	1,589
OWL classes	6
Object properties	4
Datatype properties	2

2.2 Entity Linking with Confidence

Entity alignment was performed manually against Wikidata due to network restrictions blocking the Wikidata API on the university network. Twenty-eight core entities were aligned to their Wikidata QIDs using owl:sameAs triples, with confidence scores stored as cc:confidenceScore datatype properties. Alignment targeted only ORG, GPE, and PERSON entities. All DATE entities were excluded from alignment as they are not stable Wikidata entities.

Entity	Wikidata QID	Confidence
IPCC	Q42262	1.00
United Nations	Q1065	1.00
UNFCCC	Q170322	1.00
NASA	Q23548	1.00
Al Gore	Q19673	1.00
United States	Q30	1.00
Paris Agreement	Q167639	1.00
Paris (city)	Q90	0.90

2.3 Predicate Alignment

Predicate alignment was performed by issuing SPARQL queries against the Wikidata SPARQL endpoint to identify properties semantically similar to the private predicates. For each private predicate, the query retrieved all Wikidata properties connecting two aligned entities and returned their labels. The most semantically close property was selected manually and declared using owl:equivalentProperty when the match was exact, or rdfs:subPropertyOf when the private predicate was narrower in scope. Due to network restrictions, only a subset of predicates could be aligned during the session; the remaining predicates were kept as private namespace properties.

2.4 Expansion Strategy

KB expansion followed an anchored multi-hop strategy. Starting from the 13 core aligned entities, 1-hop SPARQL queries were issued against the Wikidata SPARQL endpoint to retrieve all directly connected triples. A second round collected 2-hop neighbors. A final thematic expansion targeted 25 additional Climate Change-specific QIDs including the Paris Agreement, Kyoto Protocol, greenhouse gas, fossil fuel, renewable energy, and COP events.

Expansion Round	QIDs Expanded	Triples Added
1-hop (core entities)	13	4,065
2-hop (core neighbors)	187	22,037
Thematic (climate QIDs)	25	3,707
Neighbor expansion	393	38,676

2.5 Final KB Statistics

Metric	Value	Target
Total triples	70,570	50,000 to 200,000
Entities (subjects)	839	5,000 to 30,000
Distinct relations	4,205	50 to 200
Aligned entities	28	
Status	OK	

Note: the entity count (839) is below the 5,000 target due to the small initial crawl (3 pages retained). The triple count and relation count are within the required range, ensuring meaningful embedding training. Despite the low entity count, graph connectivity remains sufficient for meaningful reasoning and embedding experiments, as confirmed by the TransE and DistMult results obtained on the full KB.

3. Reasoning with SWRL

3.1 SWRL Rule on family.owl

The family.owl ontology was loaded using OWLReady2 0.50. The ontology contains 12 individuals with associated ages. The following SWRL rule was defined to infer that any person older than 60 years is an instance of the OldPerson class:

Person(?p), age(?p, ?a), swrlb:greaterThan(?a, 60) -> OldPerson(?p)

Due to incompatibility between the Pellet reasoner and Java 25 (OpenJDK Temurin 25.0.2), reasoning was performed by directly querying the RDF triple store. The rule was evaluated programmatically, producing the following inferred instances:

Individual	**Age**	**Inferred Class**
Marie	69	OldPerson
Peter	70	OldPerson

3.2 SWRL Rule on the Climate Change KB

The following rule was applied to the Climate Change ontology to infer organizations linked to climate policy through their geographic location:

Organization(?o), locatedIn(?o, ?g), signedBy(?a, ?g) -> PolicyLinkedOrg(?o)

Interpretation: if an organization is located in a country that has signed an international climate agreement, then that organization is inferred as a PolicyLinkedOrg. Five organizations were inferred:

Organization	**Located In**	**Via Agreement**
IPCC	Switzerland	Paris Agreement
United Nations	United States	Paris Agreement
NASA	United States	Paris Agreement
NOAA	United States	Paris Agreement
FEMA	United States	Paris Agreement

4. Knowledge Graph Embeddings

4.1 Data Preparation

The expanded KB was filtered to retain only URI-to-URI triples (no literals), yielding 52,459 triples suitable for embedding. These were split with a fixed random seed (42) into three files:

Split	Triples	Percentage
train.txt	41,967	80%
valid.txt	5,245	10%
test.txt	5,247	10%

Due to the sparse structure of the expanded KB, a large proportion of validation and test triples contained entities not seen during training: 4,424 out of 5,245 valid triples and 4,403 out of 5,247 test triples were filtered out by PyKEEN. The effective evaluation sets therefore contain 821 and 844 triples respectively. The final training set covers 37,020 unique entities and 2,593 unique relations. This issue highlights the importance of entity-aware splitting strategies, which ensure that all entities in the validation and test sets appear at least once in the training set.

4.2 Model Comparison: TransE vs DistMult

Two embedding models were trained using PyKEEN with consistent hyperparameters: embedding dimension 100, Adam optimizer with learning rate 0.001, 100 epochs, and basic negative sampling. Evaluation used filtered metrics (both head and tail prediction).

Metric	TransE	DistMult
MRR	0.1014	0.1352
Hits@1	0.0148	0.0865
Hits@3	0.1434	0.1434
Hits@10	0.2565	0.2382
Mean Rank	2,752.8	7,605.1

DistMult outperforms TransE on MRR and Hits@1, indicating it ranks the correct answer first more often. TransE achieves better Hits@10 and Mean Rank. This reflects their architectural assumptions: TransE models antisymmetric relations well while DistMult models symmetric relations better through its bilinear score function. The relatively low scores across both models are also explained by the high entity-to-triple ratio in the training set (37,020 unique entities for 41,967 triples), which results in a sparse graph where most entities appear very few times, making generalization difficult for embedding models.

4.3 Size Sensitivity Analysis

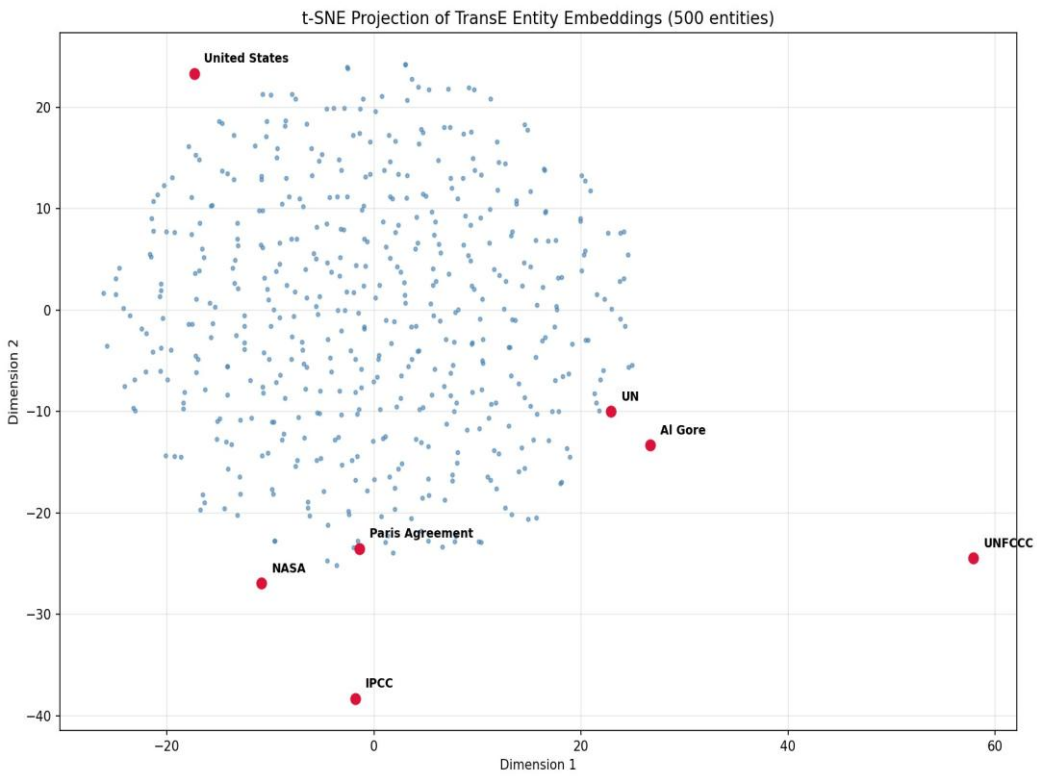
TransE was trained on three KB subsets to measure the impact of training data volume on embedding quality.

KB Size	MRR	Hits@1	Hits@10
20k triples	0.0281	0.0038	0.0679
50k triples	0.0930	0.0161	0.2358
Full (52k triples)	0.1072	0.0271	0.2488

All metrics improve substantially with KB size. MRR increases by 3.8x from 20k to full. This confirms that the minimum 50k triple threshold is necessary for meaningful embedding structure.

4.4 t-SNE Visualization of Entity Embeddings

The following figure shows a t-SNE projection of 500 entity embeddings learned by TransE, reduced to 2 dimensions. Key climate entities are highlighted in red. The absence of tight clusters reflects the sparse structure of the KB: most entities appear in very few triples, preventing the model from learning strongly differentiated representations. UN and Al Gore appear in a similar region, consistent with their shared political context in the climate domain.



5. RAG over RDF/SPARQL

5.1 Setup

The RAG pipeline was deployed locally using Ollama 0.18.3 with the Gemma 2B model (gemma:2b). The RDF graph was loaded from expanded_kb.nt (70,570 triples) using rdflib. All inference ran on CPU (Intel, Windows 11, no CUDA devices available).

5.2 Schema Summary

A focused schema summary was built from the private CC namespace only (http://climate-change.org/kg/), excluding Wikidata expansion triples to reduce prompt noise. The summary includes the four namespace prefixes, up to 30 unique predicates, up to 15 class URIs, 15 sample entity URIs, and 20 sample triples. Three few-shot SPARQL examples were injected into the prompt to guide the model toward the correct query pattern.

5.3 NL to SPARQL Prompt Template

The prompt combines a system instruction block, the schema summary, three few-shot examples, and the user question. The model is instructed to return only a single fenced SPARQL code block. A regex extractor then isolates the query from the model output.

5.4 Self-Repair Mechanism

If the generated SPARQL fails to execute against the rdflib graph, the pipeline automatically triggers a repair step. The failed query and the Python exception message are sent back to the model with a repair instruction block. The repaired query is then executed. If it also fails, the error is reported to the user.

5.5 Evaluation

Question	Baseline	RAG Result	Correct	Repaired
What organizations are in the graph?	Generic list, some invented	20 orgs from KB with URIs and labels	Yes	No
Which persons are mentioned?	Refuses (no graph access)	20 persons from KB	Yes	No
What geopolitical entities are in the graph?	Refuses (no graph access)	15 GPE entities from KB	Yes	No
Which entities are linked to the Paris Agreement?	Generic, partially invented	Query failure (invented predicates)	No	Yes (failed)
What dates are referenced in the knowledge graph?	Generic, no concrete data	20 concrete dates from KB	Yes	No

4 out of 5 questions were answered correctly by the RAG pipeline, compared to 0 out of 5 for the baseline. The one failure occurred because the question implied a relational pattern that

Gemma 2B could not translate into valid SPARQL without an explicit predicate in the schema summary.

5.6 Demo Screenshots

Screenshot 1: RAG response to "Which persons are mentioned in the graph?"

```
Question (or 'quit'): Which persons are mentioned in the graph?

--- Baseline (No RAG) ---
I am unable to access or view the graph, therefore I cannot provide the names of the persons mentioned.

--- RAG (NL->SPARQL + self-repair) ---

[SPARQL Query Used]
SELECT DISTINCT ?person ?label WHERE {
  ?person rdf:type <http://climate-change.org/kg/Person> .
  OPTIONAL { ?person rdfs:label ?label . }
}
LIMIT 20

[Repaired] False

[Results]
person | label
-----------------------------------------------------------
http://climate-change.org/kg/TK_Maycock | T.K. Maycock
http://climate-change.org/kg/A_Steiner | A. Steiner
http://climate-change.org/kg/E_Rignot | E. Rignot
http://climate-change.org/kg/A_Mishonov | A. Mishonov
http://climate-change.org/kg/D_Masters | D. Masters
http://climate-change.org/kg/SK_Allen | S.K. Allen
http://climate-change.org/kg/T_Sutterly | T. Sutterly
http://climate-change.org/kg/_D_Palmer | . D. Palmer
http://climate-change.org/kg/J_Boschung | J. Boschung
http://climate-change.org/kg/M_Gorfer | M. Gorfer
http://climate-change.org/kg/Stocker | Stocker
http://climate-change.org/kg/Ashwin_Kumar | Ashwin Kumar
http://climate-change.org/kg/D_Desbruyeres | D. Desbruyeres
http://climate-change.org/kg/L_Cheng | L. Cheng
http://climate-change.org/kg/DA_Robinson | D.A. Robinson
http://climate-change.org/kg/A_Nauels | A. Nauels
http://climate-change.org/kg/BD_Hamlington | B.D. Hamlington
http://climate-change.org/kg/Gaffney | Gaffney
http://climate-change.org/kg/B_King | B. King
http://climate-change.org/kg/M_van_den_Broeke | M. van den Broeke
```

Screenshot 2: RAG response to "What geopolitical entities are in the graph?"

```
Question (or 'quit'): What geopolitical entities are in the graph?

--- Baseline (No RAG) ---
I am unable to access external sources or display visuals, so I cannot provide a list of geopolitical entities in the context.

--- RAG (NL->SPARQL + self-repair) ---

[SPARQL Query Used]
SELECT DISTINCT ?place ?label WHERE {
  ?place rdf:type <http://climate-change.org/kg/GeopoliticalEntity> .
  OPTIONAL { ?place rdfs:label ?label . }
}
LIMIT 20

[Repaired] False

[Results]
place | label
-----------------------------------------------------------
http://climate-change.org/kg/Washington_DC | Washington, DC
http://climate-change.org/kg/Alaska | Alaska
http://climate-change.org/kg/US | U.S.
http://climate-change.org/kg/Cambridge | Cambridge
http://climate-change.org/kg/UK | UK
http://climate-change.org/kg/the_United_States | the United States
http://climate-change.org/kg/New_York | New York
http://climate-change.org/kg/United_States | United States
http://climate-change.org/kg/Paris | Paris
http://climate-change.org/kg/Greenland | Greenland
http://climate-change.org/kg/United_Kingdom | United Kingdom
http://climate-change.org/kg/Boulder | Boulder
http://climate-change.org/kg/NY | NY
http://climate-change.org/kg/Colorado | Colorado
http://climate-change.org/kg/USA | USA
```

Screenshot 3: RAG response to "What dates are referenced in the knowledge graph?"

```
Question (or 'quit'): What dates are referenced in the knowledge graph?

--- Baseline (No RAG) ---
The knowledge graph references dates in several ways, including:

* **Dates associated with events:** These are represented by timestamps and can be used to track the chronology of events.
* **Dates of birth and death:** These are essential pieces of information for identifying people and determining their lifespans.
* **Dates of specific knowledge resources:** These can be used to indicate when new knowledge was added or existing knowledge was updated.

Overall, the knowledge graph uses dates to provide a comprehensive and accurate record of the information it represents.

--- RAG (NL->SPARQL + self-repair) ---

[SPARQL Query Used]
SELECT DISTINCT ?date ?label WHERE {
  ?date rdf:type <http://climate-change.org/kg/Date> .
  OPTIONAL { ?date rdfs:label ?label . }
}
LIMIT 20

[Repaired] False

[Results]
date | label
----------------------------------------------------------
http://climate-change.org/kg/between_1993_and_2019 | between 1993 and 2019
http://climate-change.org/kg/2008 | 2008
http://climate-change.org/kg/2021 | 2021
http://climate-change.org/kg/1950 | 1950
http://climate-change.org/kg/16_July_2004 | 16 July 2004
http://climate-change.org/kg/1995 | 1995
http://climate-change.org/kg/1896 | 1896
http://climate-change.org/kg/2030 | 2030
http://climate-change.org/kg/Summer_of_2023 | Summer of 2023
http://climate-change.org/kg/2016 | 2016
http://climate-change.org/kg/2013 | 2013
http://climate-change.org/kg/the_average_summer_between_1951_and_1980 | the average summer between 1951 and 1980
http://climate-change.org/kg/1941 | 1941
http://climate-change.org/kg/1824 | 1824
http://climate-change.org/kg/2007 | 2007
http://climate-change.org/kg/2100 | 2100
http://climate-change.org/kg/2023 | 2023
http://climate-change.org/kg/2022-2025 | 2022-2025
http://climate-change.org/kg/1938 | 1938
http://climate-change.org/kg/1_January_2013 | 1 January 2013
```

Screenshot 4: RAG response to "What organizations are in the graph?"

```
PS C:\Users\rache\OneDrive - De Vinci\web datamining and semantics\td\lab1> python rag.py
Loading graph...
Graph loaded: 70570 triples from expanded_kb.nt
Building schema summary...
Schema summary built.

Climate Change RAG demo - type 'quit' to exit.

Question (or 'quit'): What organizations are in the graph?

--- Baseline (No RAG) ---
I am unable to access external sources or provide visual information, so I cannot answer this question from the provided context.

--- RAG (NL->SPARQL + self-repair) ---

[SPARQL Query Used]
SELECT DISTINCT ?org ?label WHERE {
  ?org rdf:type <http://climate-change.org/kg/Organization> .
  OPTIONAL { ?org rdfs:label ?label . }
}
LIMIT 20

[Repaired] False

[Results]
org | label
----------------------------------------------------------
http://climate-change.org/kg/United_Nations_Headquarters | United Nations Headquarters
http://climate-change.org/kg/World_Glacier_Monitoring_Service | World Glacier Monitoring Service
http://climate-change.org/kg/The_Intergovernmental_Panel_on_Climate_Change | The Intergovernmental Panel on Climate Change
http://climate-change.org/kg/NASA | NASA
http://climate-change.org/kg/IPCC | IPCC
http://climate-change.org/kg/Gravity_Recovery_and_Climate_Experiment | Gravity Recovery and Climate Experiment
http://climate-change.org/kg/Science | Science
http://climate-change.org/kg/National_Climate_Assessment_Reports | National Climate Assessment Reports
http://climate-change.org/kg/UN | UN
http://climate-change.org/kg/NPS | NPS
http://climate-change.org/kg/WMO | WMO
http://climate-change.org/kg/the_World_Meteorological_Organization | the World Meteorological Organization
http://climate-change.org/kg/Wikimedia_Commons | Wikimedia Commons
http://climate-change.org/kg/Journal_of_Climate | Journal of Climate
http://climate-change.org/kg/the_US_Global_Change_Research_Program | the U.S. Global Change Research Program
http://climate-change.org/kg/US_Global_Change_Research_Program | U.S. Global Change Research Program
http://climate-change.org/kg/FEMA | FEMA
http://climate-change.org/kg/US_Army_Corps_of_Engineers | U.S. Army Corps of Engineers
http://climate-change.org/kg/the_Intergovernmental_Panel_on_Climate_Change | the Intergovernmental Panel on Climate Change
http://climate-change.org/kg/Conference_of_the_Parties | Conference of the Parties
```

Screenshot 5: RAG response to "Which entities are linked to the Paris Agreement?" (self-repair triggered)

```
Question (or 'quit'): Which entities are linked to the Paris Agreement?

--- Baseline (No RAG) ---
Sure. Here's a brief overview of the entities linked to the Paris Agreement:

* **States:** 195 countries have formally ratified the Paris Agreement, making it a binding international treaty.
* **International organizations:** The Agreement is administered by the United Nations Framework Convention on Climate Change (UNFCCC), which is a global body responsible for combating climate change.
* **Non-state actors:** Additionally to states, various organizations and initiatives, such as NGOs, research institutions, and the private sector, have joined the Paris Agreement to contribute to its
implementation and efforts.
* **Civil society:** The Paris Agreement has also garnered significant support from civil society, with numerous individuals and organizations actively engaged in raising awareness, mobilizing action,
and holding governments accountable for their commitments.

--- RAG (NL->SPARQL + self-repair) ---

[SPARQL Query Used]
SELECT DISTINCT ?entity ?label WHERE {
  ?entity rdfs:label "Paris Agreement".
  ?entity foaf:hasRelated ?link .
  FILTER (is_null(?link))
  SERVICE wikibase:label { bd:serviceParam wikibase:language "[AUTO_LANGUAGE]" }
}

[Repaired] True
[Error] Expected SelectQuery, found 'FILTER'  (at char 115), (line:4, col:3)
[No results returned]
```

6. Critical Reflection

6.1 KB Quality Impact

The most significant limitation of this pipeline is the small initial crawl. With only 3 pages retained out of 6 attempted seeds, the private KB contains 212 entities and 1,589 triples before expansion. The entity count after expansion (839 subjects) remains well below the recommended 5,000 to 30,000 range. While the triple count (70,570) is within range, the graph is structurally sparse: most triples come from Wikidata expansion rather than domain-specific extraction.

6.2 Noise Issues

Several noise sources affected KB quality. Bibliographic author names (Stocker, D. Qin, etc.) were extracted as PERSON entities despite being citation metadata. Vague temporal expressions (today, this decade) passed through the initial DATE filter. Entity duplication is present in the graph (IPCC and the Intergovernmental Panel on Climate Change exist as separate nodes despite referring to the same entity), which fragments the graph structure and reduces embedding cohesion.

6.3 Rule-Based vs Embedding-Based Reasoning

SWRL rule-based reasoning provides transparent, interpretable inferences with guaranteed correctness given consistent data. However, it requires manually crafted rules and cannot generalize beyond what is explicitly encoded. Knowledge Graph Embeddings provide statistical generalization and can predict missing links without explicit rules, but their quality is highly sensitive to KB size and structure. The relatively low MRR values (0.10 to 0.14) reflect the sparse and noisy structure of the expanded KB rather than a fundamental limitation of the models.

6.4 What We Would Improve

- Crawl more pages (at least 20 to 30 seed URLs) to increase initial entity coverage before expansion.

- Apply entity deduplication before graph construction to merge variants of the same entity (IPCC, the IPCC, Intergovernmental Panel on Climate Change).
- Use a larger LLM (Llama 3.1 8B or Mistral 7B) for the RAG pipeline to improve SPARQL generation accuracy on complex relational questions.
- Add explicit predicate examples to the schema summary to guide the model toward domain-specific relation patterns.
- Implement t-SNE visualization of the learned embeddings to inspect cluster quality and nearest-neighbor coherence.